Introduction to logistic regression

Warmup

Work on the warmup activity (handout).

Warmup

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -19.21 + 5.45\ \mathrm{GPA}$$

What is the estimated odds of acceptance for a student with a GPA of 4.0?

Warmup

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -19.21 + 5.45\ \text{GPA}$$

> What is the estimated probability of acceptance for a student with a GPA of 4.0?

Warmup

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -19.21 + 5.45\ \mathrm{GPA}$$

Calculate the odds ratio comparing odds of acceptance for a student with a GPA of 4.0 to a student with a GPA of 3.0.

Warmup

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -19.21 + 5.45\ \mathrm{GPA}$$

> Calculate the odds ratio comparing odds of acceptance for a student with a GPA of 4.0 to a student with a GPA of 3.0.

$$\frac{\exp\{-19.21 + 5.45 \cdot 4\}}{\exp\{-19.21 + 5.45 \cdot 3\}} = 232.76$$

So the odds of acceptance are 232.76 times higher for a student with a GPA of 4.0, compared to a student with a GPA of 3.0

Warmup

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -19.21 + 5.45\ \mathrm{GPA}$$

> Calculate the odds ratio comparing odds of acceptance for a student with a GPA of 3.8 to a student with a GPA of 2.8.

Warmup

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -19.21 + 5.45 \text{ GPA}$$

> Calculate the odds ratio comparing odds of acceptance for a student with a GPA of 3.8 to a student with a GPA of 2.8.

$$\frac{\exp\{-19.21 + 5.45 \cdot 3.8\}}{\exp\{-19.21 + 5.45 \cdot 2.8\}} = 232.76$$

So the odds of acceptance are 232.76 times higher for a student with a GPA of 3.8, compared to a student with a GPA of 2.8

> This is the same odds ratio as comparing a 4.0 GPA to a 3.0 GPA!

Warmup

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -19.21 + 5.45 \text{ GPA}$$

Odds ratio for increasing GPA by one point:

Interpreting coefficients

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = \widehat{\beta}_0 + \widehat{\beta}_1 x$$

Interpretation of $\widehat{\beta}_1$:

- A one unit increase in x is associated with an increase of $\widehat{\beta}_1$ in the log odds
- A one unit increase in x is associated with an increase in the odds by a factor of $\exp\{\widehat{\beta}_1\}$

> How do you think we interpret $\widehat{\beta}_0$?

Interpreting coefficients

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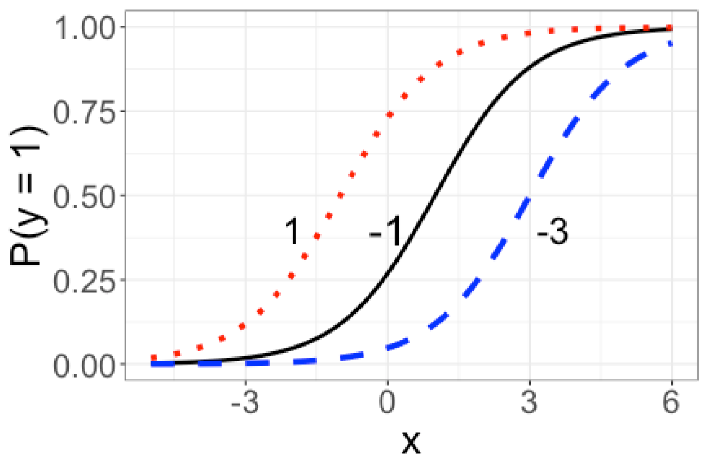
Interpretation of $\widehat{\beta}_0$:

- The estimated log odds when $x = 0$ are $\widehat{\beta}_0$
- The estimated odds when $x = 0$ are $\exp\{\widehat{\beta}_0\}$

Shape of the fitted curve

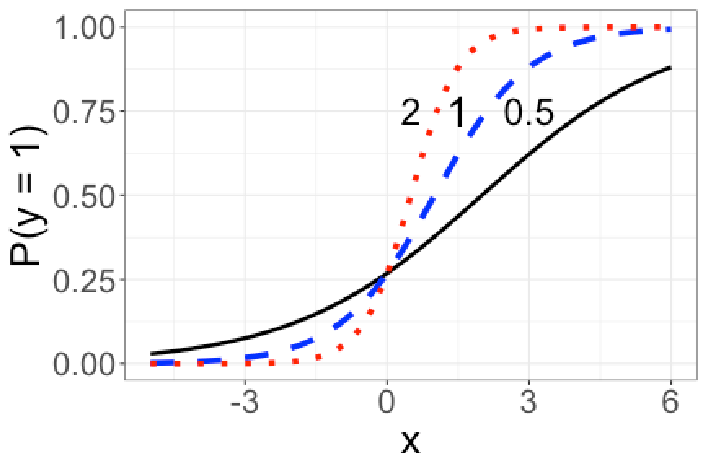
$$\widehat{\pi} = \frac{\exp\{\widehat{\beta}_0 + x\}}{1 + \exp\{\widehat{\beta}_0 + x\}}$$

$$\widehat{\beta}_0 = -3, -1, 1$$



$$\widehat{\pi} = \frac{\exp\{-1 + \widehat{\beta}_1\, x\}}{1 + \exp\{-1 + \widehat{\beta}_1\, x\}}$$

$$\widehat{\beta}_1 = 0.5, 1, 2$$



Fitting logistic regression in R

```
library(Stat2Data)
data("MedGPA")

med_glm <- glm(Acceptance ~ GPA, data = MedGPA,
               family = binomial)
```

- `glm` stands for "Generalized Linear Model"
- Use `family = binomial` for binary responses

Fitting logistic regression in R

```r
library(Stat2Data)
data("MedGPA")

med_glm <- glm(Acceptance ~ GPA, data = MedGPA,
               family = binomial)
summary(med_glm)
```

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -19.207      5.629  -3.412 0.000644 ***
GPA            5.454      1.579   3.454 0.000553 ***
```

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -19.21 + 5.45\,\text{GPA}$$

Class activity

Work on the class activity on the course website. Submit your HTML on Canvas at the end of class.

Class activity

$$\pi = P(Accepted = 1)$$

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -8.712 + 0.246 \text{ MCAT}$$

> What is the change in the odds of acceptance associated with an increase of 1 point on the MCAT?

Class activity

$$\pi = P(Accepted = 1)$$

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -8.712 + 0.246 \text{ MCAT}$$

> What is the change in the odds of acceptance associated with an increase of 1 point on the MCAT?

Answer: An increase of 1 point on the MCAT is associated with an increase in the odds of acceptance by a factor of $\exp\{0.246\} = 1.28$.

Class activity

$$\pi = P(Accepted = 1)$$

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -8.712 + 0.246 \text{ MCAT}$$

> What is the estimated probability that a student with an MCAT score of 40 is accepted?

Class activity

$$\pi = P(Accepted = 1)$$

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -8.712 + 0.246 \text{ MCAT}$$

> What is the estimated probability that a student with an MCAT score of 40 is accepted?

Answer: $\widehat{\pi} = \dfrac{\exp\{-8.712 + 0.246 \cdot 40\}}{1 + \exp\{-8.712 + 0.246 \cdot 40\}} = 0.76$

So we estimate that a student with an MCAT score of 40 has a 76% chance of being accepted to medical school.

Class activity

$$\pi = P(Accepted = 1)$$

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -8.712 + 0.246\ \mathrm{MCAT}$$

> For approximately what MCAT score would a student have a roughly 50-50 chance of being accepted to medical school?

Class activity

$$\pi = P(Accepted = 1)$$

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -8.712 + 0.246\ \text{MCAT}$$

> For approximately what MCAT score would a student have a roughly 50-50 chance of being accepted to medical school?

Answer: We need odds = 1, so

$$\frac{\widehat{\pi}}{1-\widehat{\pi}} = \exp\{-8.712 + 0.246\ \text{MCAT}\} = 1$$

So $-8.712 + 0.246\ \text{MCAT} = 0$, which means $\text{MCAT} = 8.712/0.246 = 35.4$